

Applied NLP

DATA 641

Lecture 6 — Text Classification Pipeline

*Practical Natural Language Processing: A Comprehensive Guide to Building Real-World NLP Systems 1st Edition, Sowmya Vajjala, Bodhisattwa Majumder, Anuj Gupta, Harshit Surana, O' Reilly.

Types of Machine Learning Systems—Supervised learning

In supervised learning, the training data you feed to the algorithm includes the desired solutions, called labels.

A typical supervised learning task is classification. The spam filter is a good example of this: it is trained with many example emails along with their class (spam or ham), and it must learn how to classify new emails.

Another typical task is to predict a target numeric value, such as the price of a car, given a set of features (mileage, age, brand, etc.) called predictors. This sort of task is called regression. To train the system, you need to give it many examples of cars, including both their predictors and their labels (i.e., their prices).

Here are some of the most important supervised learning algorithms

- k -Nearest Neighbors
- Linear Regression
- Logistic Regression
- Support Vector Machines (SVMs)
- Decision Trees and Random Forests
- Neural Networks

Types of Machine Learning Systems—Unsupervised learning

In unsupervised learning the training data is unlabeled.

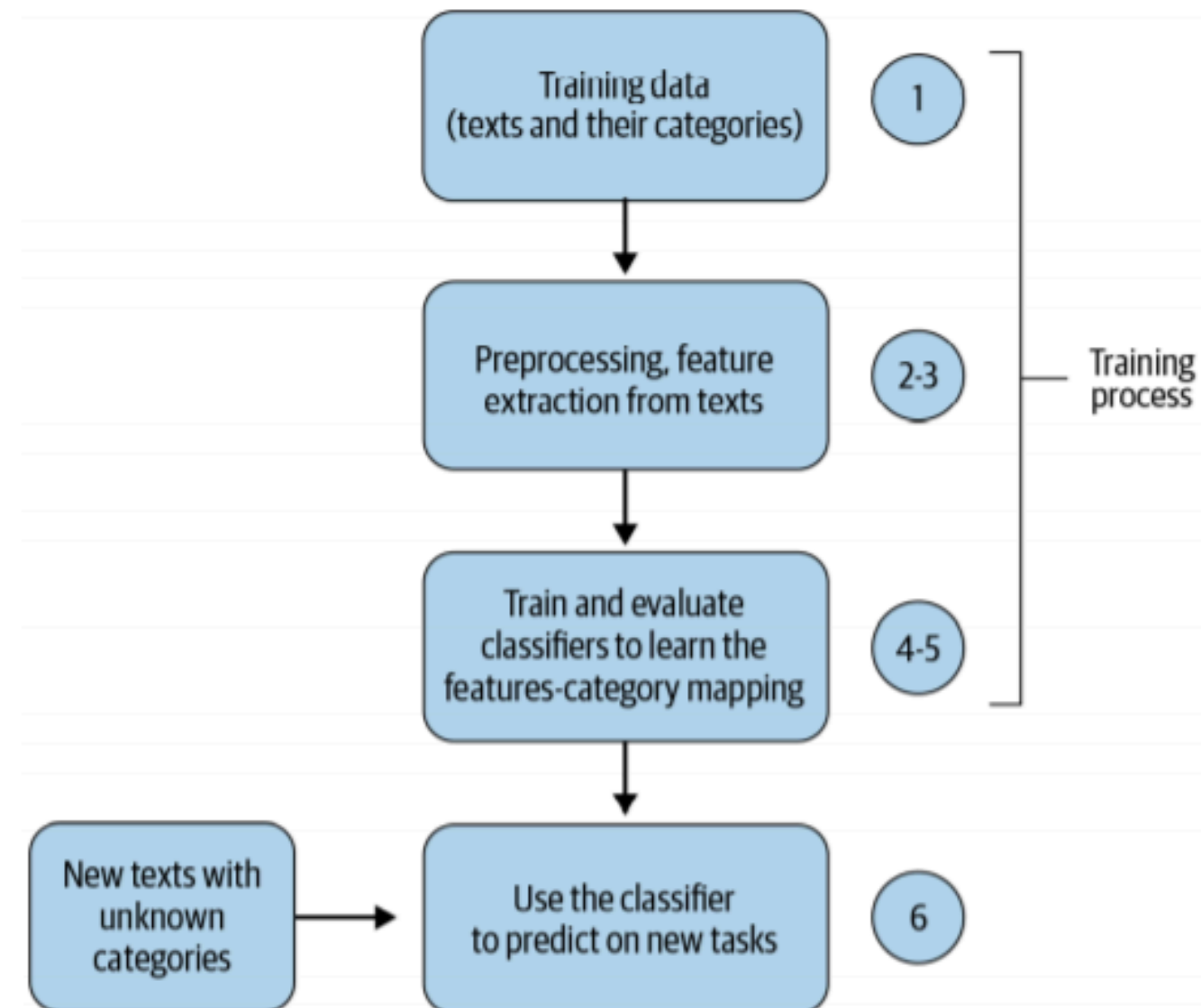
Here are some of the most important unsupervised learning algorithms

- Clustering
 - K -Means
 - Hierarchical Cluster Analysis (HCA)
- Anomaly detection and novelty detection
- Dimensionality reduction and latent variable modeling
 - Principal Component Analysis (PCA)
 - Kernel PCA
 - t-distributed Stochastic Neighbor Embedding (t-SNE)
 - Independent component analysis
 - Local Dirichlet Allocation (LDA)
 - Tensor factorization

An overview of text classification

- Text classification is an NLP task that involves categorizing or assigning predefined labels or categories to textual documents or snippets.
- It is a fundamental problem in NLP and has applications in various domains such as sentiment analysis, spam detection, topic classification, and document categorization.
- Text classification algorithms aim to automatically learn patterns, features, or representations from the text data to make predictions or assign appropriate labels.
- It involves training a machine learning or deep learning model on a labeled dataset, where the input is a textual document and the output is a class or category label.

Text classification systems a flowchart



Some applications include

- Content classification and organization
- Misinformation detection
- Authorship attribution
- Customer support



GoRedForWomen  @GoRedForWomen · Mar 7
Our **#WCW** belongs to Lilly Rocha, Karen Hill and all the women who have survived heart disease and stroke. We are proud to stand with you in building healthier lives. **#RedDressCollection** presented by **@Macys**

Gwen Stefani  @gwenstefani · 22 Nov 2017
Get in the holiday spirit & watch me perform on the **@Macys** Thanksgiving Day Parade tomorrow on **@NBC**! 🎉 **#YouMakeltFeelLikeChristmas** ❄️
#MacysParade

Some applications include

- E-commerce



Reviewed 21 February 2013

Good food service lacking

The cafe in the Jordaan is lovely in a cold Winter day or outside in the summer. Food and drinks great and a very good menu.

Service though is abysmal. There is a fine line between over eager and good service but here I had to get up and ask for everything I needed. Shame but still worth the visit.